

CSMFAB Master Cost Analysis & Market Viability Report

Carrington Storm Motors — Safe Pod Engineering Company

All Products: Ranked by Per-Unit Cost | Seed Money Strategy | June 2026

Document Type: Cumulative Cost Analysis + Commercial Viability Assessment **Scope:** All 43 fabricated products across CSMFAB, CSMIC, CSMMECH, CSMPARK, CSMMetal **Pricing Basis:** 2026 material market rates; production-scale COGS (5,000-50,000 units/year) **Purpose:** Identify lowest-cost-to-enter products for immediate market revenue to fund full portfolio fabrication

Executive Summary

The CSMFAB portfolio spans 43 distinct products across 9 commercial categories. Per-unit production costs range from **\$32** (BFRP rebar, per 6m bar) to **\$108,000** (Knott's Berry Farm per 100m² module). The full portfolio cannot be launched simultaneously. A strategic staged approach — beginning with **12 immediately marketable consumer products** — can generate seed revenue of an estimated **\$8.5M–\$22M in Year 1** at projected initial volumes, funding tooling and material supply chains for the full product line.

Critical cost finding: MXene Ti₃C₂T_x at \$2,000/kg (2026) is the single highest per-unit cost driver for 18 products. Price projections of \$400-600/kg by 2028-2029 as production scales will dramatically improve margins across the entire portfolio.

Part I — All Products Ranked: Least to Most Expensive (Production Volume COGS)

Production Volume = 5,000-50,000 units/year; fully tooled

Rank	Product	Part #	Category	Prod COGS	MSRP Range	Gross Margin
1	BFRP Rebar (per 6m bar)	009	Structural	\$32	\$55–\$75	42-57%
2	Non-Conductive PEN Spray Can (per 12-pack)	018	Hardware	\$58	\$95–\$130	39-55%
3	Ulnar-Rest CLD	036	Embark	\$180	\$280–\$380	36-53%
4	Armrest Thorax-Calm STF Seatbelt Set (4-belt)	033	Embark	\$200	\$300–\$420	33-52%
5	Ancient Hearth Stone Warmer	008	Kitchen	\$210	\$330–\$480	36-56%
6	Vibra-Volt Pedal Cover Set	026	Embark	\$275	\$420–\$580	35-53%

Rank	Product	Part #	Category	Prod COGS	MSRP Range	Gross Margin
7	Pet-Safe Infrasound Cargo Liner	037	Embark	\$295	\$460–\$620	36-52%
8	Stellar-Tint EV Window Film (per vehicle)	034	Embark	\$295	\$460–\$620	36-53%
9	Fluid-Damp Gear Shift Interface	035	Embark	\$200	\$320–\$440	38-54%
10	Passive EMF Detector	021	Safety	\$235	\$360–\$500	35-53%
11	Dielectric Rope & Rigging Kit (50m)	017	Hardware	\$265	\$420–\$580	37-54%
12	Way-Finder Haptic Insoles (pair)	038	Embark	\$420	\$640–\$860	34-51%
13	Aegis Aerogel Thermal Break Panel (m ²)	003	Insulation	\$380	\$600–\$820	37-54%
14	Silence- Block LRAM Door Insert Set	032	Embark	\$880	\$1,380– \$1,800	36-51%
15	Aegis Geopolymer Concrete w/ ZrB ₂ (m ³)	107	Infra	\$910*	\$1,200– \$1,600	24-43%*
16	Non- Conductive Plumbing System	012	Plumbing	\$650	\$1,000– \$1,350	35-52%
17	Inductive- Proof Wiring System	013	Electrical	\$720	\$1,100– \$1,500	34-52%
18	Bio-Sync Seat Cushion	MAG	Embark	\$440	\$680–\$920	35-52%
19	Neural-Grip MRE Steering Wheel Wrap	029	Embark	\$540	\$820–\$1,100	34-51%
20	Grounding- Sole Footwear (pair)	024	Embark	\$650	\$1,000– \$1,350	35-52%

Rank	Product	Part #	Category	Prod COGS	MSRP Range	Gross Margin
21	Cervical-Guard	030	Embark	\$490	\$760–\$1,020	36-52%
22	Headrest Safe Stove	014	Kitchen	\$620	\$960–\$1,300	35-52%
23	BFRP Range Hood	006	Plumbing	\$480	\$740–\$1,000	35-52%
24	Ceramic Water Heater Cloud-Nest	028	Embark	\$780	\$1,200–\$1,600	35-52%
25	Aerogel Child Seat Base	002	Kitchen	\$780	\$1,200–\$1,600	35-52%
26	Ancient Hearth ZrB2 Stove Burner	020	Motors	\$1,280	\$1,900–\$2,500	33-49%
27	Non-Conductive Piezoelectric Motor	007	Kitchen	\$780	\$1,200–\$1,600	35-52%
28	Stone-Bake Geothermal Air Fryer Oven	031	Embark	\$1,580	\$2,400–\$3,200	34-51%
29	Mag-Float Diamagnetic Seat Rail	016	Hardware	\$730	\$1,120–\$1,500	35-52%
30	Non-Conductive Door Hardware Set	011	EMF Shield	\$1,180	\$1,800–\$2,400	34-51%
31	Aegis Home Building Envelope Shield Kit	013	Electrical	\$720	\$1,100–\$1,500	34-52%
32	Non-Conductive Wiring System	010	Structural	\$680	\$1,050–\$1,400	35-52%
33	Conductive-Proof Building Materials Suite Ceramic & PEEK Hand Tool Set (12pc)	015	Tools	\$1,050	\$1,600–\$2,100	34-50%

Rank	Product	Part #	Category	Prod COGS	MSRP Range	Gross Margin
34	Safe-Wave GaN Solid-State Microwave	005	Kitchen	\$980	\$1,500–\$2,000	35-51%
35	Stellar-Chill Elec-trocaloric Refrigerator	004	Kitchen	\$1,580	\$2,400–\$3,200	34-51%
36	Aegis-C Composite Shielding Panel (m ²)	001	Structural	\$1,320	\$2,000–\$2,700	34-51%
37	Aegis GFRP Pipeline (per 100m)	106	Infra	\$29,500	\$42,000–\$60,000	30-42%
38	Aegis Power Line Monopole (per pole)	105	Infra	\$4,200	\$6,500–\$8,500	36-51%
39	Stealth-Lift BFRP Forklift	019	Industrial	\$48,000	\$72,000–\$95,000	33-49%
40	Phantom Menehune Zookeeper Robot	MECH003	Robotics	\$15,000	\$25,000–\$35,000	40-57%
41	Phantom MK-1 Aegis Hardened Robot	MECH001	Robotics	\$48,000	\$80,000–\$110,000	40-56%
42	Robot Self-Destruct Chip System	MECH002	Robotics	\$11,000	\$20,000–\$28,000	45-61%
43	Aegis-Ascension Drone Backpack	050	Emergency	\$9,500	\$18,000–\$28,000	47-66%
44	Knott's Berry Farm Module (per 100m ²)	PARK	Recreation	\$108,000	\$180,000–\$250,000	40-60%

Note: Aegis Concrete base mix (no ZrB2 whiskers) = \$340/m³ COGS. With ZrB2 EMF shielding additive: \$13,800/m³.

Part II — Products Grouped by Category

A. Aegis Kitchen (6 products)

Products 002, 004, 005, 006, 007, 008 — Appliances for Carrington-safe food preparation and storage. **COGS range:** \$210–\$1,580 | **Combined household set COGS:** ~\$4,800 | **Total MSRP:** \$7,500–\$11,000

B. Aegis Building Materials (6 products)

Products 001, 003, 009, 010, 011, 107 — Wall panels, rebar, insulation, shielding, concrete. **COGS range:** \$32–\$1,320 per unit | **Lowest unit cost in portfolio:** BFRP rebar at \$32/bar

C. Aegis Home Services (3 products)

Products 012 (plumbing), 013 (wiring), 014 (range hood) — Complete home infrastructure hardening. **COGS range:** \$620–\$720 per system

D. Aegis Hand Tools & Hardware (4 products)

Products 015, 016, 017, 018 — Tools, door hardware, rope, spray cans. **COGS range:** \$58–\$1,050 | **Spray can is the volume/low-cost entry consumer product**

E. Aegis Embark — Vehicle Interior (15 products)

Products 024, 026, 028-038, MAG — Full vehicle interior ecosystem. **COGS range:** \$180–\$1,580 | **Multiple entry-price products for immediate launch**

F. Aegis Industrial (2 products)

Products 019 (forklift), 020 (motor) — Industrial GIC-immune machinery. **COGS range:** \$1,280–\$48,000

G. Infrastructure (3 products)

CSMIC 105, 106, 107 — Grid, pipeline, concrete. B2G/B2B market only. **COGS range:** \$340/m³–\$29,500/100m^{**}

H. Robotics (3 products)

CSMMECH 001, 002, 003 — Humanoid, self-destruct, swarm robots. **COGS range:** \$11,000–\$48,000

I. Emergency & Recreation (2 products)

CSMFAB 050 (drone), CSMPARK 001 — Specialty/institutional. **COGS range:** \$9,500–\$108,000/module

Part III — Shared Material Infrastructure (Cost-Sharing Opportunities)

These materials are used across multiple products. Establishing supply contracts early dramatically reduces per-unit costs across the entire portfolio:

Material	Used In (# products)	Current 2026 Cost	Projected 2028–2030 Cost	Annual Volume at Full Portfolio
Basalt fiber (Elium® BFRP)	28 products	\$5–12/kg	\$4–9/kg	45,000 kg/yr
Elium® thermoplastic resin	28 products	\$12/kg	\$9/kg	28,000 kg/yr
KNbO3-BaTiO3 piezo	18 products	\$180/kg	\$90/kg	800 kg/yr
YInMn Blue pigment	35 products	\$75/kg	\$55/kg	3,200 kg/yr
MXene Ti3C2Tx	18 products	\$2,000/kg	\$400-600/kg (2028)	120 kg/yr
LORD MRF-140CG	8 products	\$250/kg	\$200/kg	2,400 kg/yr
Polyimide-silica aerogel	14 products	\$250/m ²	\$160/m ²	4,800 m ² /yr
Si3N4 bearings + gear	12 products	\$90/kg	\$70/kg	1,200 kg/yr
PVDF-TrFE film	10 products	\$350/kg	\$220/kg	600 kg/yr

Portfolio supply contract strategy: Establish framework agreements with: - Elium®: Arkema SA (France) — anchor customer for US distribution - MXene: Drexel University spin-off / Murata (Japan production partner) - Basalt fiber: Mafic USA (Ireland/US) — US domestic supplier - YInMn Blue: Shepherd Color Company (Cincinnati, OH)

Part IV — Market Viability Analysis: Seed Money Strategy

Tier 1 — IMMEDIATE LAUNCH (0-12 months, minimal tooling NRE)

Products ready with <\$50,000 NRE, clear market, existing distribution channels

Product	Part #	Est. Year 1 Units	Revenue (MSRP)	Profit Contribution
BFRP Rebar (per bar)	009	50,000 bars	\$2.75M–\$3.75M	\$1.15M–\$2.15M
PEN Spray Can (12-packs)	018	8,000 packs	\$760K–\$1.04M	\$320K–\$540K
Passive EMF Detector	021	3,500 units	\$1.26M–\$1.75M	\$455K–\$840K
Dielectric Rope Kit	017	2,000 kits	\$840K–\$1.16M	\$310K–\$580K
Stellar-Tint Window Film	034	5,000 vehicles	\$2.3M–\$3.1M	\$825K–\$1.63M
Ulnar-Rest Armrest	036	4,000 units	\$1.12M–\$1.52M	\$400K–\$680K
Thorax-Calm Seatbelt Set	033	3,000 sets	\$900K–\$1.26M	\$300K–\$580K
TIER 1 TOTALS		75,500 sales units	\$9.93M–\$13.58M	\$3.78M–\$7.77M

Year 1 Seed Revenue: \$3.8M–\$7.8M profit contribution Required startup capital for Tier 1: \$385,000 NRE + \$1.2M working capital

Tier 2 — NEAR-TERM LAUNCH (12-24 months, moderate NRE \$50K-\$500K)

Products requiring dedicated tooling but addressing clear premium markets

Product	Part #	NRE Required	Year 2 Revenue Projection
Grounding-Sole Footwear	024	\$180,000	\$3.5M–\$5.4M
Vibra-Volt Pedal Cover	026	\$95,000	\$2.1M–\$3.2M
Cervical-Guard Headrest	030	\$120,000	\$2.8M–\$4.2M
Neural-Grip Wheel Wrap	029	\$95,000	\$3.1M–\$4.6M
Ancient Hearth Stone Warmer	008	\$60,000	\$1.4M–\$2.2M
Inflatable Water Heater	006	\$75,000	\$2.2M–\$3.4M
Safe Stove Range Hood	014	\$190,000	\$2.8M–\$4.2M
TIER 2 TOTALS		\$815,000 NRE	\$17.9M–\$27.2M

Tier 3 — MEDIUM-TERM LAUNCH (24-48 months, \$500K–\$5M NRE)

Full Aegis Home kitchen suite, Aegis Embark complete line, Industrial products

Products 001–005, 011–013, full Embark suite (028-038), industrial 019–020 **Estimated cumulative NRE for Tier 3: \$4.5M–\$9.5M Year 3-4 revenue projection: \$45M–\$85M/year at scale**

Tier 4 — LONG-TERM / B2G (48+ months, \$10M+ infrastructure)

Products 050 (drone), 105-107 (infrastructure), CSMMECH 001-003 (robots), CSMPARK **Market:** Government procurement, utility contracts, defense **Estimated market entry: \$150M–\$400M/year at maturity**

Part V — Recommended 5-Product Seed Money Bundle

Based on lowest NRE, fastest time-to-market, best margins, and clearest distribution path:

□ **Product 1: BFRP Rebar**

Why: Zero exotic equipment; pull-winding is standard. Existing steel rebar market = \$26B/yr. Immediate structural contractor channel. Mark: **\$32 COGS □ \$65 MSRP = 51% margin.**

□ **Product 2: Passive EMF Detector**

Why: No moving parts, no exotic processes except chromatic glaze. First-mover in \$0 existing market. Consumer + industrial safety segment. Privacy + space weather monitoring trend. Mark: **\$235 COGS □ \$420 MSRP = 44% margin.**

□ **Product 3: Stellar-Tint EV Window Film**

Why: Roll-to-roll manufacturing; existing window film infrastructure can be contracted. 7M+ new EVs/year in US alone. NIR + circadian + EMI in one film = unique value prop. Mark: **\$295 COGS □ \$520 MSRP = 43% margin.**

□ **Product 4: Dielectric Rope & Rigging Kit**

Why: UHMWPE rope is standard; ZrO2/PEEK hardware is precision machined but small volumes. Immediate industrial electrical safety + marine markets. Mark: **\$265 COGS □ \$480 MSRP = 45% margin.**

□ **Product 5: Thorax-Calm STF Seatbelt Set**

Why: TPU and STF fluid are low-cost; PEEK buckle covers are standard machined parts. Automotive aftermarket + fleet safety segment. Clear liability reduction story. Mark: **\$200 COGS □ \$380 MSRP = 47% margin.**

Combined 5-Product Seed Bundle — Year 1 Analysis

Total required startup capital: \$280,000 (NRE) + \$850,000 (working capital) = **\$1.13M** **Conservative Year 1 revenue at 50% of projections:** \$5.8M **Conservative Year 1 profit:** \$2.4M **ROI on seed capital:** ~213% in Year 1

This \$2.4M profit, reinvested, fully funds Tier 2 NRE (\$815,000) with \$1.6M remaining for materials supply contracts (Arkema Elium®, Mafic basalt fiber, Shepherd YInMn Blue).

Part VI — Cost Trajectory: MXene Price Drop Impact

The single largest future cost improvement across the portfolio:

Year	MXene Price (/kg)	Products Impacted	Average COGS Reduction
2026 (current)	\$2,000	18 products	Baseline
2027	\$1,200	18 products	8-15% avg. reduction
2028	\$600	18 products	18-32% avg. reduction
2029	\$350	18 products	25-42% avg. reduction
2030	\$180	18 products	30-50% avg. reduction

By 2030, 18 products will see COGS reductions of 30-50% purely from MXene price normalization. This makes the full portfolio significantly more competitive as the decade advances — and aligns with the Solar Cycle 25/26 elevated risk timeline driving consumer awareness and demand.

Part VII — Revenue to Full Portfolio Funding Map

SEED CAPITAL (\$1.13M)

□

Year 1: 5 Seed Products □ \$2.4M Profit

□

Year 2: + 7 Tier 2 Products (NRE \$815K fully funded)

Revenue grows to \$12M-\$18M

□

Year 3: + Full Aegis Kitchen Suite + Full Embark Line (NRE \$4.5M-\$9.5M funded)

Revenue grows to \$38M-\$65M

□

Year 4: Infrastructure contracts (B2G, utility) + Robot deployments

Revenue: \$80M-\$140M

□

Year 5: Full portfolio operational

Projected annual revenue: \$150M-\$280M

Total capital required for full portfolio (across 5 years): \$14M-\$25M Funded entirely from portfolio revenue — no external dilutive funding required after Year 2

End of CSMFAB Master Cost Analysis & Market Viability Report Carrington Storm Motors | Safe Pod Engineering Company | June 2026 All cost figures are estimates based on 2026 materials market and comparable manufacturing benchmarks. Actual costs will vary by supplier selection, production facility, and volume negotiation.